(144)

Ex: 21:- 462, 506, 552, 600, 650, ?

Ans:- here in the question the numbers are looks nearby squares.

like:- $462 = 441 + 21 = 21^2 + 21$ (or) $462 = 484 - 22 = 22^2 - 22$.

$506 = 484 + 22 = 22^2 + 22$ (or) $506 = 529 - 23 = 23^2 - 23$

$552 = 529 + 23 = 23^2 + 23$ (or) $552 = 576 - 24 = 24^2 - 24$

$600 = 576 + 24 = 24^2 + 24$ (or) $600 = 625 - 25 = 25^2 - 25$

$650 = 625 + 25 = 25^2 + 25$ (or) $650 = 676 - 26 = 26^2 - 26$

$\therefore$ Next No. Qs $\Rightarrow 26^2 + 26 = 676 + 26 = 702 = 27^2 - 27 = 729 - 27$.

logic:- $n^2 + n$ / $n^2 - n$.

* Note:- Skeletons for square series Qs:-

n^2, $n^2 \pm$ constant / odd / even / prime / square / cubes / ! / tables ..... n (given square).

(vi) Cube series:- for cube series we must know cubes from 1 to 15.

* Note:- Skeletons for cube series:-

n^3, $n^3 \pm$ constant / odd / even / Prime / square / cube / ! / powers / tables ... n.

$n^3 \pm n^2$, $n^3 \pm n^2 \pm$

(vii) Miscelaneous Series:- [Puzzle Series]:- Its a series generally not looks like a series for the first observation. if we observe keenly & practice more of it then only we can able to recognize the logic behind it.

Ex:-22:- 3,3,5,4,4,3,5,5,4,3,6,6,8, ?

Ans:- Most of the students think that this Q is a series of 2 common terms, different term (3,3,5) but it is applicable till 3,3,5,4,4,3 only, 5,5,4 only but from 4,3,... this Q is different. So, this Q is different.

Logic:- number of letters in spellings of natural numbers from 1 to 13 given.

∴ Answer is no. of letters in spelling of fourteen = 8

one = 3 letters, two = 3 letters, three = 5 - - - - - - - - - - - thirteen = 8, Fourteen = 8.

Ex:-23:- 1, 11, 21, 1211, 111221, 312211, ?

Ans:- We have to write given observations in terms of words then we can get the logic.

first observation = 1 = one time one = one one = 11

2nd observation = one one = 11.

3rd observation = two times one [11] = two one = 21

4th obs = one time two one time one [21] = one two one one = 1211

5th obs = [one one][one two][two one][1211] = 111221.

6th obs = [three one][two two][one one] = 312211.

7th obs = [one three][one one][two two][two one] = 13112221.

In this example first no. of times & then which digit is taken is considered.

1 → one time one → 11 → two times one → 21

one time: No. of times; one: which digit. two times: No. of times; one: which digit.